Asymptotic counting of D(n)-tuples

Problèmes ouverts — A. Dujella · 2026

Problem 1.7 (Dujella, 2026). Let $D_{m,n}(N)$ denote the number of $D(n)$-m-tuples with elements in the set $\{1, 2, \ldots, N\}$ for a given positive integer N.

(a) If n is a perfect square, is there a constant $C(n)$ depending only on n, such that $D_{3,n} \sim C(n)N \log N$? Is it true that $C(n) = C(1) = \frac{3}{\pi^2}$?

(b) If n is not a perfect square, is there a constant $C(n)$ depending only on n, such that $D_{3,n} \sim C(n)N$?

(c) If n is a perfect square, what can be said about the asymptotic behaviour of $D_{4,n}$?

(d) If n is not a perfect square, is it true that $D_{4,n}$ is finite ?

Remark. An affirmative answer to question (b) in the case when $|n|$ is prime or $n = -1$ is given in Adžaga, N., Dražić, G., Dujella, A., Pethő, A. : Asymptotics of $D(q)$-pairs and triples via L-functions of Dirichlet characters. Ramanujan J. **66**, Article 11 (2025).

Questions (a) and (b) are completely solved in Badesa, J. : On the asymptotics of $D(n)$-pairs and triples. Ramanujan J. **67**, Article 101 (2025). The answers to both questions are affirmative. In particular, for n a perfect square, $D_{3,n} \sim \frac{3}{\pi^2} N \log N$.

An alternative proof of (a) and (b) is given in the recent paper Dražić, G., Kazalicki, M., Mrazović, R. : Equidistribution of Diophantine pairs among the equivalence classes of quadratic forms. Preprint (2025). arXiv : 2512.22902.